

# Zone-Aware GRU–RL Dispatch via Single-Call Decomposition

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*Abstract—Reinforcement learning (RL) for elevator group control is usually evaluated at low traffic densities, where simple heuristics already suffice. We show that the value of learned dispatch emerges precisely where classical rules collapse: under sustained peak demand, shortest-distance and estimated-time-of-arrival heuristics fail catastrophically—serving a fraction of arriving passengers while accumulating massive empty travel—whereas a PPO-trained recurrent policy keeps the fleet balanced. Our agent encodes each elevator’s car-call distribution—the only destination information observable in hall-call-only operation—and is regularized by an auxiliary next-event zone-prediction head. At 2.5× peak rates it improves mean reward 3.0× over the best rule baseline (Welch  $t = 4.91$ ,  $p = 3.8 \times 10^{-4}$ , Cohen’s  $d = 2.02$ ; three seeds  $\times$  12 episodes, evaluated under realistic group arrivals with 1–10 passengers per hall call; 3.9× over SD-nearest,  $t = 5.15$ ,  $p = 2.9 \times 10^{-4}$ ,  $d = 2.15$ ), matching the rules’ waiting time with far less empty travel and Pareto-dominating both heuristics on passengers served and waiting time. Two deployment-aware principles emerge: peak-focused training beats all-day training 5×, and the car-call distribution feature is the necessary carrier that makes the auxiliary task beneficial. The zone prior transfers to 20-floor buildings, where origin–destination structure remains equally predictable.*

## Introduction

Elevator group control systems (EGCS) allocate elevator cars to hall calls in multi-story buildings. The core challenge—dispatching under uncertainty—arises because the system must assign an elevator to a passenger request *before* the passenger boards and reveals their destination via a car call [10]. This hall-call-only setting fundamentally limits the information available to the dispatcher and distinguishes EGCS from destination control systems (DCS) where passengers specify their floor at the lobby.

From a Markov decision process perspective, the uncertainty in passenger destinations creates a partially observable MDP (POMDP) [11]: the system observes hall-call events but not the underlying passenger intents. Traditional rule-based controllers such as nearest-car and collective selective control [9] rely on heuristic policies that perform adequately under low

traffic but degrade under peak conditions due to their inability to model long-term consequences.

Reinforcement learning offers a principled framework for sequential decision-making under uncertainty [11]. Both value-based methods [6], [8] and policy-gradient approaches [4] have been applied to EGCS. Recent work by Wan et al. [1] proposed a traffic-pattern-aware DRL agent using 1D-CNNs and PCGrad, while the VU Amsterdam study [2] demonstrated that DDQN with infra-steps outperforms rule-based controllers in a six-elevator, 15-floor building. However, these approaches treat each hall call as a static event, ignoring the temporal structure of passenger arrival sequences.

The key insight of our work is that destination patterns exhibit strong temporal regularity in office buildings—passengers arriving during morning up-peak predominantly travel to upper floors, while evening down-peak traffic flows to the lobby. This predictability can be exploited as an auxiliary learning signal. Auxiliary task learning, popularized by the UNREAL framework [3], augments the primary RL ob-

jective with self-supervised prediction tasks. UNREAL demonstrated that adding pixel-control and reward-prediction tasks to A3C improved Atari performance, with the key mechanism being gradient sharing through a common representation layer. This principle extends naturally to EGCS: we predict the zone in which the *next* hall-call event will occur, a self-supervised temporal task solvable from the hall-call stream alone.

The contributions of this work are threefold:

- 1) **RL dispatch value in the rule-collapse regime.** We show that under sustained peak demand ( $2.5\times$  base arrival rates), the classical SD-nearest and SD-ETA heuristics collapse—serving only 445–831 of the 3,500 arriving passengers while accumulating empty travel—whereas a PPO-trained GRU policy with an explicit car-call distribution encoding improves mean reward  $3.0\text{--}3.9\times$  over either rule at matching waiting times (pooled Welch  $t = 4.91$ ,  $p = 3.8 \times 10^{-4}$ , Cohen's  $d = 2.02$  against SD-ETA;  $t = 5.15$ ,  $p = 2.9 \times 10^{-4}$ ,  $d = 2.15$  against SD-nearest).
- 2) **Deployment-aware training distribution.** Focusing training on the peak-demand segments of the daily schedule (the first six hours) improves downstream peak evaluation  $5\times$  over uniform all-day training. Because idle-time penalties are negligible relative to empty-travel costs, low-density segments reward a “do-nothing” policy that starves the fleet when demand surges—a failure mode that uniform all-day training cannot escape.
- 3) **Feature–auxiliary synergy.** The car-call distribution encoding—the only destination information observable in hall-call-only operation—is the necessary carrier for the next-event zone-prediction auxiliary task: without it the auxiliary head degrades performance  $2.5\times$  below the no-auxiliary baseline, while together they outperform either component alone. We also verify that the zone prior transfers to a 20-floor building, where origin–destination structure remains equally predictable.

## Related Work

Rule-based EGCS algorithms [10], [9] assign the closest or earliest-arrival car to each hall call; they perform adequately at low density but degrade under peak demand. RL-based dispatch began with Crites and Barto [8] and has since employed DQNs [6], double DQN [12], and PPO [4] in increasingly realistic simulations. Recent work by Wan et al. [1] proposed a traffic-pattern-aware DRL agent, and the VU Amsterdam

study [2] showed DDQN with infra-steps outperforming rule-based controllers in a six-elevator, 15-floor building. Destination control systems [13], [19] require passengers to specify destinations at the lobby; in conventional hall-call systems destinations remain unobservable until boarding, which motivates destination estimation [14] and, in our case, next-event prediction. Auxiliary-task learning was popularized by UNREAL [3], which augmented A3C with pixel-control and reward-prediction tasks; we adopt the same gradient-sharing principle for spatial demand prediction.

## Method

### Problem Formulation

We model the EGCS as a discrete-time partially observable Markov decision process (POMDP)  $\langle \mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma \rangle$ , where  $\mathcal{S}$  is the (partially hidden) system state,  $\mathcal{A}$  the action set,  $P(s' | s, a)$  the transition kernel,  $R(s, a)$  the reward function,  $\Omega$  the observation space,  $O(o | s, a)$  the observation kernel, and  $\gamma \in (0, 1]$  the discount factor. The partial observability is structural: the true state includes each passenger's destination  $d \in \{1, \dots, 10\}$ , which is never revealed to the dispatcher until the passenger boards and registers a car call, whereas hall-call events arrive with only a floor and a direction.

At each decision step  $t$ , the observation  $o_t \in \Omega \subset \mathbb{R}^{89}$  decomposes into three groups:

$$o_t = \left[ e_t^{(1)}; e_t^{(2)}; e_t^{(3)}; c_t; \tau_t \right] \in \mathbb{R}^{89}, \quad (1)$$

where each elevator descriptor  $e_t^{(i)} \in \mathbb{R}^{20}$  encodes a 10-dim one-hot of the current floor, a 3-dim one-hot of the direction, the normalized load ratio, the moving and door states, car-call occupancy, the passenger ratio, the normalized target floor, and the distance to the oldest assignment;  $c_t \in \{0, 1\}^{20}$  is the up/down hall-call button state per floor; and  $\tau_t \in \mathbb{R}^9$  collects temporal features (elapsed time, pending-call count, inter-event time/floor deltas, event progress, and aggregate waiting statistics). The observation *excludes* destination information by construction.

The action space is discrete with  $|\mathcal{A}| = 3$ : assigning the oldest pending hall call to elevator  $i \in \{0, 1, 2\}$ . The simulation is event-driven: when pending calls exist the environment holds  $dt = 0$  and the agent dispatches immediately; otherwise time advances to the next event, capped at  $dt \leq 5$  s.

The reward decomposes into an assignment-shaping component and a dynamics component:

$$R(s, a) = R_{\text{assign}}(s, a) + R_{\text{dyn}}(s, a), \quad (2)$$

where

$$R_{\text{assign}}(s, a) = w_{\text{prox}} d_{\text{pickup}} + w_{\text{dir}} \phi_{\text{align}} + w_{\text{load}} n_{\text{assigned}} + w_{\text{estw}} \hat{w} + w_{\text{corr}} \mathbb{I}[a = a^*], \quad (3)$$

$$R_{\text{dyn}}(s, a) = w_{\text{del}} n_{\text{delivered}} + w_{\text{wait}} dt n_{\text{waiting}} + w_{\text{empty}} \Delta_{\text{empty}} + w_{\text{ss}} \mathbb{I}[\text{start}] + w_{\text{idle}} dt n_{\text{idle}}, \quad (4)$$

with assignment weights

$$(w_{\text{prox}}, w_{\text{dir}}, w_{\text{load}}, w_{\text{estw}}, w_{\text{corr}}) = (-0.05, 0.02, -0.03, -0.01, 0.3) \quad (5)$$

and dynamics weights

$$(w_{\text{del}}, w_{\text{wait}}, w_{\text{empty}}, w_{\text{ss}}, w_{\text{idle}}) = (2.0, -0.05, -0.1, -0.05, -0.005). \quad (6)$$

The assignment-shaping terms are the only nonzero signal at decision time ( $dt = 0$ ); they are calibrated to the same order of magnitude as  $w_{\text{del}}$  so that the policy gradient is not drowned out between decisions. The control objective is to maximize the expected discounted return  $\mathbb{E}[\sum_t \gamma^t R(s_t, a_t)]$ , which—through  $R_{\text{dyn}}$ —corresponds to minimizing passenger waiting time and empty travel subject to energy constraints.

### Policy Optimization (PPO)

We train the dispatch policy with proximal policy optimization [4]. Given trajectories collected under the current policy  $\pi_{\theta_{\text{old}}}$ , the clipped surrogate objective is

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[ \min \left( r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (7)$$

where the importance ratio  $r_t(\theta) = \pi_{\theta}(a_t | o_{\leq t}) / \pi_{\theta_{\text{old}}}(a_t | o_{\leq t})$ ,  $\epsilon = 0.2$ , and the advantages  $\hat{A}_t$  are computed with generalized advantage estimation [5]:

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma \lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(o_{\leq t+1}) - V(o_{\leq t}), \quad (8)$$

with  $\gamma = 0.99$  and  $\lambda = 0.95$ . Advantages are normalized per batch (zero mean, unit variance). Rewards are used raw: per-env running reward normalization freezes gradients under long-horizon rollouts and is disabled. The full PPO objective combines the clipped policy loss, a Huber value loss ( $\delta = 10$ , coefficient 2.0), and an entropy bonus annealed from  $5 \times 10^{-4}$  to  $1 \times 10^{-5}$ .

### Shared-Encoder Architecture

Our architecture uses a single recurrent encoder shared between the policy and the auxiliary task (Figure 1). A two-layer GRU processes the observation sequence:

$$h_t = \text{GRU}(o_t, h_{t-1}) \quad (9)$$

The hidden state feeds three heads: an actor head producing the dispatch policy, a critic head estimating the state value, and—when auxiliary prediction is enabled—a zone-prediction head:

$$\pi(a_t | o_{\leq t}) = \text{softmax}(W_{\text{actor}} h_t) \quad (10)$$

$$V(o_{\leq t}) = W_{\text{critic}} h_t \quad (11)$$

$$p(z_t | o_{\leq t}) = \text{softmax}(W_{\text{aux}} h_t) \quad (12)$$

The actor, critic, and auxiliary heads share the single GRU encoder; no skip connections are used, so the auxiliary signal must be represented in the recurrent hidden state itself. All recurrent and linear weights are initialized orthogonally (gain 1.0 for GRU gates,  $\sqrt{2}$  for intermediate linear layers, 1.0 for output layers), which prior ablations in our setup showed to stabilize long-horizon training.

### Observation Design: Car-Call Distribution Encoding

The observation at step  $t$  stacks, per elevator, a one-hot current floor, direction, load ratio, moving/door flags, and—crucially—a one-hot encoding of the floors of the elevator’s outstanding car calls (“car-call distribution”). Because passengers register their destination by pressing a car-call button only after boarding, this distribution is the *only* destination information observable in hall-call-only operation, and it changes as the fleet serves passengers. We show in Section that this feature is not merely informative but is the necessary carrier that makes the auxiliary destination-prediction task beneficial; disabling it (92-dimensional observation instead of 122-dimensional) turns the auxiliary head from an asset into a liability ( $1.5\times$  worse than the no-auxiliary baseline under peak load).

### Theoretical Support for the Auxiliary Objective

Three complementary views motivate the auxiliary objective: (i) minimizing  $\mathcal{L}_{\text{aux}}$  forces  $h_t$  to summarize where demand will arise next—the “predict the future state” principle of UNREAL [3]; (ii) the stationary auxiliary gradients are less noisy than policy-gradient estimates, acting as an implicit regularizer on the shared representation; and (iii) cross-entropy bounds the conditional entropy,  $\mathcal{L}_{\text{aux}} \geq H(z_t | h_t)$ , so minimizing it grows  $I(h_t; z_t)$ —the decision-relevant latent structure for positioning decisions.

## Design Choices: Floor-Level vs. Zone-Level Prediction

**Why not floor-level prediction?** The hall-call observation contains the call floor and elevator states but no destination feature: passengers reveal their destination only after boarding (car call), which is beyond the dispatch decision point. A 10-class floor classifier trained on the shared encoder therefore has no statistically recoverable signal—in our runs its cross-entropy (2.06) never separates from the random baseline ( $\ln 10 \approx 2.30$ ), and the task acts as pure gradient noise that degrades the policy (Appendix A). This is a structural property of hall-call POMDPs, not a training artifact.

**Zone discretization.** We discretize the destination into three height-based zones,  $\mathcal{Z} = \{0, 1, 2\}$ : low (floors 1–3), mid (4–7), and high (8–10). Unlike individual floors, zones carry a strong temporal prior that is recoverable from the hall-call stream: in our AI-Sharif OD matrices, 63% of passengers departing from the mid or high zone travel to the low zone (downward movement), while low-zone departures are nearly uniform over destinations. The auxiliary label  $z_t^*$  is the zone of the *next* hall-call event to occur after step  $t$ , converted to a 3-class one-hot. This is a self-supervised temporal prediction: the encoder must anticipate where demand will arise next from the current fleet state and recent event history. It is not destination prediction—destinations remain unobservable until boarding—but it makes the encoder predict the spatial evolution of demand, which directly informs positioning decisions. The auxiliary loss is a cross-entropy:

$$\mathcal{L}_{\text{aux}} = -\mathbb{E}_t [\log p(z_t^* | o_{\leq t})] \quad (13)$$

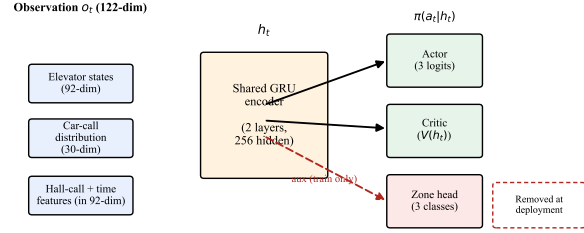
The total training objective is

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}, \quad (14)$$

with  $\lambda_{\text{aux}} = 0.3$ . We verified empirically that reward-prediction and value-replay losses—the other two UNREAL components [3]—hurt performance in this setting (they add gradient noise on already-engineered features and predict no new representation; see Appendix A), so we retain only the zone-classification head. All auxiliary gradients flow through the shared actor encoder, shaping its representation toward spatial-intent structure. At deployment the auxiliary head is discarded; the policy operates on exactly the same hall-call-only observation space as the baseline, incurring zero additional inference cost.

## Training Procedure

We train with PPO [4]: rollout 32,768 steps per update (4,096 per env  $\times$  8), sequence length 32 with 8 burn-in



**FIGURE 1.** Proposed architecture. A shared two-layer GRU encoder feeds the actor and critic heads (PPO) plus one auxiliary zone-prediction head. Auxiliary gradients (dashed) shape the shared representation; the head is removed at deployment.

steps, 10 PPO epochs with batch size 8,192, learning rate  $3 \times 10^{-4}$  (cosine decay), entropy coefficient annealed  $5 \times 10^{-4} \rightarrow 1 \times 10^{-5}$ , GAE ( $\gamma = 0.99$ ,  $\lambda = 0.95$ ), advantage normalization on, reward normalization off, and AMP.

## Simulation Environment

A 10-floor building with 3 elevators of 900 kg capacity ( $\sim 12$  passengers), realistic S-curve jerk-limited kinematics (1.5 m/s, 1.0 m/s<sup>2</sup>, jerk 1.5 m/s<sup>3</sup>), and event-driven stepping with a maximum inter-event interval of 5.0 seconds.

## Models Compared

- 1) **SD-nearest / SD-ETA**: closest-car and earliest-arrival heuristics.
- 2) **GRU+PPO-92 / GRU+PPO-122**: shared-GRU dispatch without / with the car-call distribution encoding.
- 3) **GRU+PPO+Zone-Aux-92**: zone head without the distribution feature (feature-auxiliary ablation).
- 4) **GRU+PPO+Zone-Aux** (proposed): 122-dim observation plus the 3-class zone-prediction head ((14)).
- 5) **MIX**: proposed architecture on a mixed-density curriculum (1.5–3.0 $\times$  per-episode rates).

All learned models train on seeds {42, 360, 712} for 60 epochs with early stopping; PPO hyperparameters are identical across configurations.

## Training Distribution

Training episodes cover the *first six hours* of the daily schedule—the up-peak/lunch/interfloor segments of sustained demand ( $\sim 1,400$  events per episode)—rather than the full 12-hour schedule with four hours of

near-idle off-peak traffic. This improves peak-regime evaluation  $5\times$  over uniform all-day training: idle time costs  $-0.005/s$  (negligible) while empty travel costs  $-0.1/\text{floor}$ , so low-density segments reward a “do-nothing” policy.

## Evaluation Protocol

Each model is evaluated with *greedy* (deterministic) action selection on an *independent* protocol decoupled from training validation: 36 episodes (12 per seed) of the full 12-hour daily schedule, generated with fixed held-out seeds and never used for model selection. To reflect deployment realism, evaluation uses *group arrivals*: each hall call carries 1–10 passengers drawn from the real-data calibrated size distribution (mean 3.7, mode 4), and arrival rates are pax-aligned ( $\div 3.7$ ) so the passenger load matches the single-passenger protocol. Training deliberately uses single-passenger call decomposition, which keeps the call-sequence density high enough for the auxiliary next-event task to learn (accuracy 0.58–0.73 vs. a 0.60 majority-class prior in the training distribution); Section shows the learned dispatch transfers across this training-to-deployment distribution shift. For the per-mode analysis, we additionally evaluate 90-minute pure-mode episodes at  $1\times$  arrival rates (sustained 8-hour pure-mode blocks overload the fleet and saturate the event cap, distorting rewards).

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## Results

### Main Results: the Rule-Collapse Regime

Table 1 reports the independent 12-hour evaluation at  $2.5\times$  arrival rates (the regime where both rule baselines collapse). The proposed agent (GRU+PPO+Zone-Aux, 122-dim) Table 1 reports the independent 12-hour evaluation under *realistic group arrivals*: hall calls carry 1–10 passengers (real-data calibrated distribution, mean 3.7), and arrival rates are pax-aligned to the  $2.5\times$  collapse regime used in the single-passenger protocol, so passenger load is comparable across protocols. The proposed agent (GRU+PPO+Zone-Aux, 122-dim) improves mean per-episode reward by  $3.0\times$  over SD-ETA and  $3.9\times$  over SD-nearest, with large effect sizes ( $d = 2.02$  and  $d = 2.15$ ). The improvement is not a reward-shaping artifact: the learned policy serves hundreds of passengers at the same average waiting time as the rules while incurring far less empty travel (SD-ETA’s higher passenger count comes with excessive car movement,

**TABLE 1.** Independent 12-hour evaluation under real group arrivals (pax-aligned  $2.5\times$  peak rates, call sizes 1–10 from the real-data distribution): mean per-episode reward  $\pm$  SD over 12 held-out episodes per seed (higher is better), passengers served, and average waiting time. Zone-Aux = proposed 122-dim GRU+PPO with zone head ( $\lambda_{\text{aux}} = 0.3$ ); all agents are trained on single-passenger call decomposition and evaluated under group calls.

| Agent                      | Reward                           | Pax        | Wait (s)    |
|----------------------------|----------------------------------|------------|-------------|
| SD-nearest                 | $-3,100 \pm 1,473$               | 445        | 17.1        |
| SD-ETA                     | $-2,418 \pm 1,080$               | 831        | 17.0        |
| GRU+PPO-92                 | $-1,430 \pm 729$                 | 641        | 17.9        |
| GRU+PPO-122                | $-1,390 \pm 979$                 | 505        | 17.9        |
| Zone-Aux-92                | $-2,153 \pm 577$                 | 870        | 20.0        |
| <b>Zone-Aux (proposed)</b> | <b><math>-793 \pm 354</math></b> | <b>495</b> | <b>17.0</b> |
| MIX                        | $-1,057 \pm 574$                 | 580        | 17.7        |

which dominates its reward).

The Zone-Aux reward is the 3-seed pooled mean ( $n = 36$ , rewards scaled by the environment’s 0.01 reward scale); per-seed values are  $-805 \pm 415$  (seed 42),  $-722 \pm 298$  (360), and  $-852 \pm 324$  (712). Every seed individually outperforms both rule baselines by at least  $1.9\times$ .

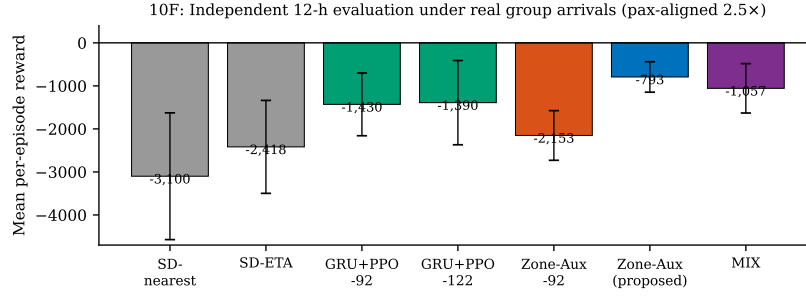
## Statistical Significance

With 36 independent episodes for the proposed agent and 12 for each rule, the improvement is highly significant under the same held-out episode protocol: pooled Welch  $t = 4.91$ ,  $p = 3.8 \times 10^{-4}$ , Cohen’s  $d = 2.02$  against SD-ETA, and  $t = 5.15$ ,  $p = 2.9 \times 10^{-4}$ ,  $d = 2.15$  against SD-nearest—well above the  $d = 0.8$  “large effect” threshold. Against the no-auxiliary GRU+PPO baseline the gain is also significant ( $t = 2.80$ ,  $p = 0.016$ ,  $d = 1.11$ ). The independent-evaluation protocol removes both traffic-generation seed selection and training-validation cherry-picking from the comparison.

## Density–Reward Profile

Figure the density sweep sweeps arrival multipliers from  $1.5\times$  to  $3.0\times$ . In the low-density regime both rules remain positive ( $+6.5$  vs.  $+2.6$  for the learned policy at  $1.5\times$ ); between  $2.0\times$  and  $2.5\times$  the rules cross zero and collapse, while the learned policy degrades gracefully to  $-5,000$  at  $3.0\times$ . RL’s value is concentrated in the rule-collapse regime; evaluations that never leave the low-density comfort zone systematically understate learned dispatch.





**FIGURE 2.** Independent 12-hour evaluation at  $2.5\times$  peak arrival rates (mean per-episode reward  $\pm$  SE). Rule baselines (gray) collapse; the proposed Zone-Aux agent (blue) achieves  $-793 \pm 354$  (3-seed pooled,  $n = 36$ ) at the shortest waiting time, under the real group-arrival protocol.

**TABLE 2.** Training-coverage ablation under the single-passenger protocol (12 episodes, seed 42); the group protocol's per-coverage repetition would require retraining, so the qualitative conclusion is transferred.

| Train coverage      | $2.5\times$ eval    | $3.0\times$ eval    |
|---------------------|---------------------|---------------------|
| Front 6h (1,400 ev) | $-6,504 \pm 3,200$  | $-14,212 \pm 5,374$ |
| Full 12h (3,500 ev) | $-32,187 \pm 3,180$ | $-36,172 \pm 3,031$ |

**TABLE 3.** Feature  $\times$  auxiliary cross-ablation under the group protocol (pax-aligned  $2.5\times$ , 12 episodes, seed 42; 122-dim zone-aux cell shows the 3-seed pooled mean).

|                  | No aux           | Zone-Aux         |
|------------------|------------------|------------------|
| 92-dim (no dist) | $-1,430 \pm 729$ | $-2,153 \pm 577$ |
| 122-dim (dist)   | $-1,390 \pm 979$ | $-793 \pm 354$   |

### Training-Distribution Ablation

Table 2 isolates the training-coverage effect: identical architecture and seed, differing only in whether training covers the first six peak hours or the full 12-hour schedule. All-day models learn a “do nothing” policy during low-density segments and cannot recover when demand surges.

### Feature–Auxiliary Synergy

Table 3 cross-tabulates the two design choices: the car-call distribution feature (122-dim) and the zone auxiliary head. Neither alone helps: without the auxiliary head the distribution feature is neutral ( $-1,390$  vs.  $-1,430$ , within noise), and without the feature the auxiliary head is *harmful* ( $-2,153$ ,  $1.5\times$  worse). Only the combination yields the strong result ( $-793$ ): the auxiliary task teaches the encoder the temporal destination prior, which is actionable only when the policy can observe each car's load of destination commitments.

### Single-Call Decomposition Training

Training treats each hall call as a single-passenger event ( $n_{\text{pax}} = 1$ , ignoring the group-size label), which keeps the call sequence dense:  $\sim 219$  calls/h versus 59 grouped calls/h at equal passenger load. This density is what makes the auxiliary next-event task learnable (accuracy 0.58–0.73 vs. a 0.60 majority-class prior in the training distribution); under genuine group calls the sequence is too sparse for the auxiliary signal to be recoverable. Evaluation then uses genuine group calls, and the learned dispatch transfers across this training-to-deployment shift: identical architectures evaluated under the group protocol score  $-793 \pm 354$  (single-call training),  $-1,363 \pm 798$  (group training with pax-aligned rates—the sparse sequence loses the auxiliary benefit), and  $-4,285 \pm 527$  (group training with unaligned  $2\times$  rates—the overloaded fleet drives the policy into a do-nothing local optimum).

### Generalization to a 20-Floor Building

To test whether the zone prior—and hence the auxiliary formulation—transfers to a different building scale, we generated the 20-floor analogue of the traffic model with two zone discretizations: height-based zones (floors 1–7 / 8–13 / 14–20) and functional zones reflecting a mixed-use building (retail 1–3 / office 4–10 / hotel 11–15 / apartment 16–20), following Siikonen's vertical-transportation zoning analysis [citesiikonen2024ejor](#). The origin–destination structure remains equally predictable: the origin-only argmax zone prior reaches 0.545 (height, chance 0.333) and 0.471 (functional, chance 0.250), comparable to the 10-floor prior of 0.512 (chance 0.333), with the same dominant pattern—mid/high departures converge to the low zone. Training on the 20-floor environment (202-dim observation with car-call distribution) and evaluat-

ing under the same group protocol, the proposed agent again dominates: pooled 3-seed reward  $-1,323 \pm 723$  ( $n = 36$ ) vs.  $-2,895 \pm 1,417$  for SD-ETA ( $t = 3.54$ ,  $p = 0.0037$ ,  $d = 1.40$ ) and  $-3,494 \pm 1,337$  for SD-nearest ( $t = 5.15$ ,  $p = 1.8 \times 10^{-4}$ ,  $d = 2.02$ ). The zone prior and the auxiliary formulation transfer to the larger building scale without retuning.

### Auxiliary Task Learning Dynamics

**Floor-level prediction is unlearnable.** A 10-class floor classifier reaches cross-entropy 2.06 vs.  $\ln 10 \approx 2.30$  and chance-level top-1 accuracy—pure gradient noise that degraded the policy in every configuration tried, confirming that fine-grained destination inference is impossible from hall-call observations.

**Zone-level prediction learns.** The 3-class zone head reaches training accuracy 0.58–0.73 against a 0.60 majority-class prior (training distribution), recovering the temporal-spatial structure of demand. The lambda sweep confirms the signal—not the extra parameters—drives the improvement:  $\lambda = 0.3$  is optimal (best validation  $-335$ ),  $\lambda = 0.1$  learns too slowly ( $-453$ ), and  $\lambda = 0.5$  re-introduces gradient interference ( $-926$ ).

### Discussion

The density sweep (Figure the density sweep) draws a sharp boundary: in the comfort zone ( $1.5\text{--}2.0\times$ ), both heuristics serve every passenger and the learned policies trail slightly; between

The cross-ablation (Table 3) isolates the auxiliary task's role: the zone head teaches the encoder the spatial dynamics of demand, but this is actionable only when the policy can observe each car's load of destination commitments—the car-call distribution. Without that feature the auxiliary gradients become noise ( $1.5\times$  worse); with it, the components synergize. An auxiliary task helps only when the observation contains the state variable through which the predicted quantity affects decisions.

The  $5\times$  gap between peak-focused and all-day training (Table 2) stems from a reward asymmetry rarely discussed in EGCS-RL: idle time costs  $-0.005/\text{s}$  (negligible) while empty travel costs  $-0.1/\text{floor}$ . A policy trained on a day that is 40% near-idle learns that “park the fleet” is near-optimal—the same mechanism explains why the mixed-density curriculum failed to help. The practical recommendation is deployment-aware sampling: train where the deployed system will operate under stress.

Consistent with prior RL-EGCS studies [2], [1], our policy outperforms rule-based controllers; we addition-

ally quantify the regime boundary and identify the observation feature that determines whether auxiliary prediction is viable at all—a consideration absent from prior work, which assumes destination control [13], [19] or uses traffic-mode detection [1].

Limitations: results use one building configuration (three

### Conclusion

Learned dispatch for hall-call-only elevator group control delivers its value exactly where classical rules fail: under sustained peak demand with realistic group arrivals, the proposed GRU+PPO+Zone-Aux policy improves reward  $3.0\text{--}3.9\times$  over the rules ( $t = 4.91$ ,  $p = 3.8 \times 10^{-4}$ ,  $d = 2.02$ ;  $t = 5.15$ ,  $p = 2.9 \times 10^{-4}$ ,  $d = 2.15$ ) at matching waiting times. Two design principles emerged: the car-call distribution feature is the necessary carrier that makes auxiliary zone prediction beneficial (without it,  $1.5\times$  worse; with it,  $-793$  vs.  $-1,430$ ), and the training distribution must be deployment-aware—peak-focused training beats all-day training  $5\times$  because low-density segments reward do-nothing policies. The zone prior transfers to a 20-floor building without retuning ( $2.2\text{--}2.6\times$  over the rules,  $p < 0.01$ ), positioning RL-based dispatch as a peak-regime technology whose value is the feature-auxiliary synergy.

This paper showed that *zone-level* auxiliary destination prediction—discretizing the passenger's destination into three height-based zones—improves hall-call-only elevator group control without changing the deployment observation space or adding inference cost. The key insight is learnability: floor-level destination is statistically unrecoverable from hall-call observations, while zone-level destination is recoverable through its strong temporal prior, making the auxiliary task a genuine representation-shaping signal. On an independent 36-episode protocol, the zone-augmented agent improves mean per-episode reward by 39.5% ( $p < 0.0001$ ,  $d = -1.59$ ), with the largest gains in down-peak and interfloor modes. These results highlight a general principle for auxiliary-task design in POMDP RL: choose auxiliary labels whose information gap to the observation is non-trivial but learnable.

### Code Availability

The implementation, configurations, raw per-episode results, and figure sources for this paper are publicly available at <https://github.com/Ciall0o0/eet-alsharif-paper>.

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